（別紙様式 4）

修士学位論文要旨・研究経過報告書

【芸術工学専攻】

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※日本語 1,600 字以内，英語 800 ワード以内で記述してください。（裏面への記載は不可）
なお，本様式にならってワード等の文書作成ソフトにより作成したものでも可とします。

論文題目（又は研究題目）：**A Study on Co-creation Tools for Designing Interpretive Signage—**
—Case Study of Fukuoka City Botanical Garden Greenhouse

1. Research Background and Objective

Interpretive signage functions as the primary medium for informal learning in botanical gardens, yet the field consistently registers two measurable deficits: low text read-through rates and poor information retention (Bitgood, 2013; Serrell, 2015). Field reconnaissance at Fukuoka Botanical Garden corroborated both failures at scale. Systematic literature review identified two structural causes: existing signage practice lacks evidence-based design principles, and no objective diagnostic instrument exists to evaluate interpretive effectiveness—leaving iterative improvement without empirical grounding (Spooner et al., 2024). Both failures have been documented across museum, zoo, and botanic garden contexts; yet design-science remediation protocols grounded in visitor data remain scarce. The deficits are structurally coupled: absent principled design guidance, practitioners cannot produce cognitively accessible signs; absent diagnostic tools, they cannot determine whether any intervention has succeeded. This study directly addresses both gaps by formalizing co-created design principles and operationalizing a low-cost, open-source eye-tracking instrument to quantify their effect on read-through rates.

2. Research Strategy and Overall Design

The greenhouse zone of Fukuoka Botanical Garden served as the research site. The study instantiated a bottom-up paradigm—problem identification, principle formalization, quantitative diagnosis, experimental verification—grounded in Tilden's (1957) interpretive theory, Ham's (2016) EROT model, and Serrell's (2015) label design framework. Co-creation Design was adopted as the central strategy: all interventions were derived from documented visitor pain points rather than top-down expert assumptions. The overall framework encapsulates an evidence-first logic in which each design decision is anchored to a documented visitor need.

3. Methodology

The study employed a mixed-methods design across four sequential phases.

Phase 1: Literature Review. Classical literature on interpretive system design, attention allocation, and cognitive load was synthesized to define the research scope and formalize the study architecture.

Phase 2: Co-creation Workshop and User Survey. A participatory design session at Global Goals Jam Kyushu 2025 engaged citizens, designers, and garden staff. Semi-structured field interviews generated qualitative data across diverse visitor demographics; thematic coding produced five primary pain-point categories, confirming the two core research problems and delineating specific visitor experience barriers.

Phase 3: Principle Derivation and Baseline Audit. Three principles targeting cognitive load reduction were formalized through systematic literature synthesis and expert interviews (Wang & Zhang, 2025a): (A) Inspirational Atmosphere—deploying interrogative or narrative framing to stimulate initial engagement; (R) Audience Relevance—mapping plant characteristics to visitors' sensory experiences and everyday knowledge; (S) Structural Clarity—enforcing legible information hierarchy and balanced typography.

A custom annotation instrument, Signage Annotator, was then deployed to audit all 70 existing greenhouse signs (N = 70). The audit confirmed that principles A and R were systematically absent from the current signage corpus.

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Phase 4: A/B Eye-tracking Experiment (Within-subjects). Fifteen authentic visitors (ages 20–80) were recruited for an in-situ controlled experiment between May and June 2025. Redesigned stimuli applied A/R/S principles as the sole manipulated variable against matched baseline signs. SIGN Visual Attention—a purpose-built, signage-integrated non-contact eye-tracking system—passively captured gaze heatmaps and scanpaths as visitors traversed their natural pathways, requiring no participant-worn instrumentation. Semantic Areas of Interest (AOI) analysis quantified text read-through rates at the word-region level.

4. Results and Discussion

The A/B experiment produced consistent positive effects across all 15 participants. Whole-sign text read-through rate in the initial 10 seconds increased from 38.6% to 45.3% in the intervention condition. The Audience Relevance region registered the most pronounced shift: 10-second reading rate rose from 5.0% to 21.4%, and full dwell time climbed from 8.5% to 35.1%, with every participant showing an upward trajectory. The four-fold increase in the Audience Relevance full dwell rate—from 8.5% to 35.1%—represents a magnitude rarely observed in field-based interpretive signage research. Effect decomposition identified the R principle—experiential knowledge linkage—as the decisive variable in capturing and sustaining gaze. The A principle (rhetorical framing) applied in isolation failed to generate stable fixation, establishing that co-creation interventions should prioritize experiential anchoring over linguistic embellishment.

5. Conclusion and Ph.D. Research Outlook

This study delivers three interrelated contributions to interpretive signage design. Methodologically, it delineates a framework for organizing, filtering, and translating visitor feedback into actionable design specifications. In applied terms, the A/R/S principles constitute transferable guidance for botanical gardens, science museums, and analogous informal learning environments. Instrumentally, SIGN Visual Attention—open-source, passive, and deployable without specialist infrastructure—demonstrates viability for field-based gaze research at minimal cost. Collectively, these outputs operationalize the methodological and empirical foundation for Research Problem 1; the R principle alone elevated full dwell time from 8.5% to 35.1%, constituting a practically significant gain for informal learning design.

The doctoral phase will target Research Problem 2—information retention—by integrating cognitive load theory (Sweller, 1988) and dual coding theory (Paivio, 1971). Full-session eye-tracking paired with delayed memory tests will investigate how systematic signage design variables (spatial layout, thematic zoning, and inter-sign continuity) modulate memory encoding, with the aim of resolving spatial discontinuity as a structural barrier to learning pathway formation in botanical garden contexts.

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